

Cattle-proof fencing

- **What?** Barbed wire
- **Who?** Joseph Glidden
- **Where and when?** US, 1874

Barbed wire made large-scale cattle farming more practical by massively reducing the cost of enclosing land, because it was much cheaper and easier to erect than wooden fences. American Michael Kelly invented the

basic design for barbed wire in 1868, but Joseph Glidden improved upon the idea, and patented it in 1874. He helped turn America's Great Plains into profitable farming country.

Sharp barbs keep cattle at bay.

Lightweight tractor

- **What?** The Ivel
- **Who?** Dan Albone
- **Where and when?** UK, 1903

The Ivel was created by the English manufacturer and inventor Dan Albone, and is widely considered to be the first successful tractor designed to replace horses. It was described as a lightweight, gasoline-powered agricultural motor for general purpose.



Combine harvester

- **What?** Self-propelled harvester
- **Who?** Holt Manufacturing
- **Where and when?** US, 1911

In 1836, in the US, Hiram Moore built and patented the first combine harvester—pulled by horses—capable of reaping, threshing, and winnowing grain.

In 1911, the first self-propelled harvester was produced by the Holt Manufacturing Company in California.



Crop-spraying drone

- **What?** Agras MG-1
- **Who?** DJI
- **Where and when?** China, 2015

Farmers have long used aircraft for spraying crops with pesticides, but it is expensive. "Dusting" by drone, however, is cheap and efficient. In 2015, Chinese drone company DJI exhibited a crop-spraying drone, called the Agras MG-1, which can fly for 12 minutes at a time, before needing to be refueled.

Crop-spraying drone in China, 2017



Gasoline-powered tractor

- **What?** Froelich tractor
- **Who?** John Froelich
- **Where and when?** US, 1892

In 1892, John Froelich invented a gasoline-powered farm vehicle. It was not a success, but his 1914 model was better received. Seeing its potential, the John Deere company bought Froelich's business.



Froelich's 1892 machine

Self-tie baler

- **What?** New Holland baler
- **Who?** Edwin Nolt
- **Where and when?** US, 1937

Early machines compacted hay, but farmhands had to bind the bales manually. In 1937, American farmer Edwin Nolt built a self-tying hay baler. The idea was picked up by the New Holland Machine Company and put into production.



A tractor tows a modern baler